CHIEF RED BEAR CHILDREN'S LODGE PLATFORM TECHNICAL ARCHITECTURE SPECIFICATION

Updated Comprehensive System Architecture, Domain Schemas, Component Topologies, and Handover Guide
Revision 2 — Expanded Independent Engineering Assessment

Client Organization: Chief Red Bear Children's Lodge / Cowessess First Nation, Saskatchewan, Canada
Technical Baseline: Post-Phase 5 Hardening | **Git Commit HEAD:** 0746579 | **Branch:** main
Repository Path: C:\Users\Shilley Pc\crbcl | **Remote:** https://github.com/innovatorledger-web/crbcl
Audience: Lead Architects, Technical Leadership, System Reviewers, and Independent Software Evaluators

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■■ TECHNICAL ARCHITECTURE & HANDOVER DISCLOSURE

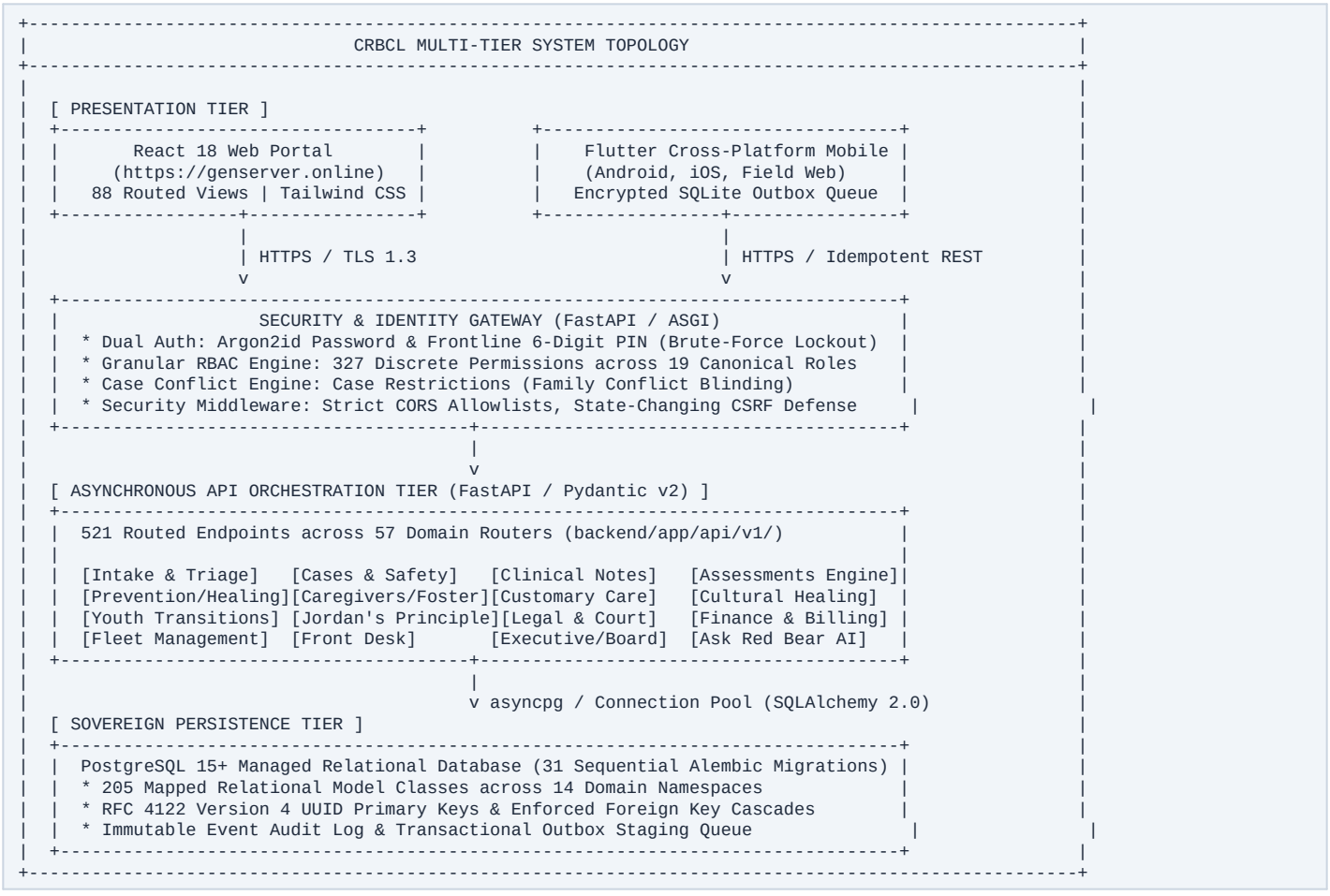
Technical Handover & Verification Notice: This document provides an exhaustive, verified engineering specification of the CRBCL software architecture as of Git commit 0746579. It is designed to support independent code review, security audits, and technical handover. While the backend architecture (205 models, 521 endpoints) and test suites are fully implemented in source code, operational production readiness requires hosting pipeline synchronization, cloud LLM credential provisioning, and frontline supervisor User Acceptance Testing (UAT).

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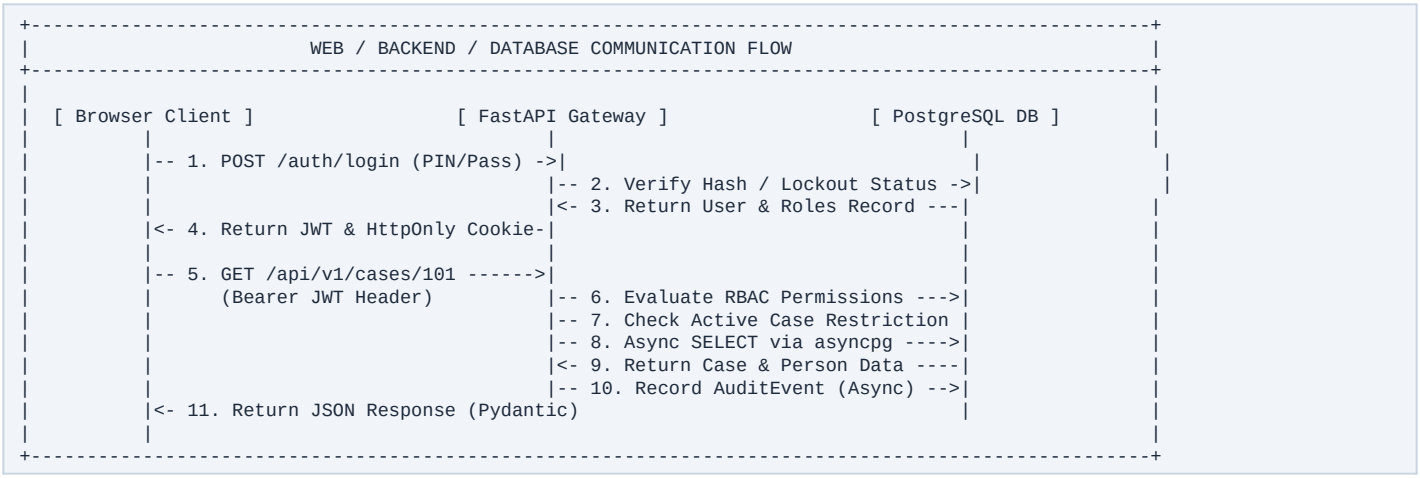
1. SYSTEM ARCHITECTURE & COMPONENT TOPOLOGY

Multi-tier system topology separating presentation, API orchestration, persistent relational storage, and background processing.



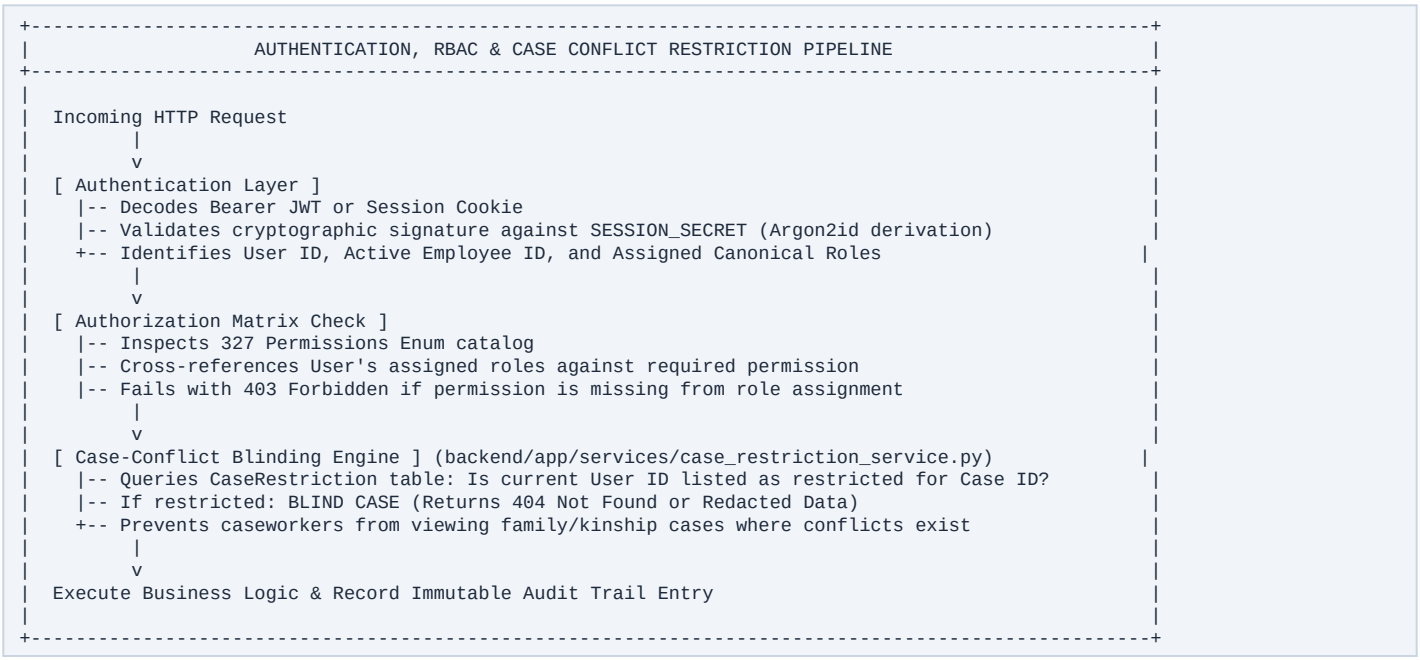
2. COMMUNICATION FLOW & API ORCHESTRATION

TLS 1.3 communication between clients and the FastAPI backend with automated JWT cycling and state-changing CSRF defense.



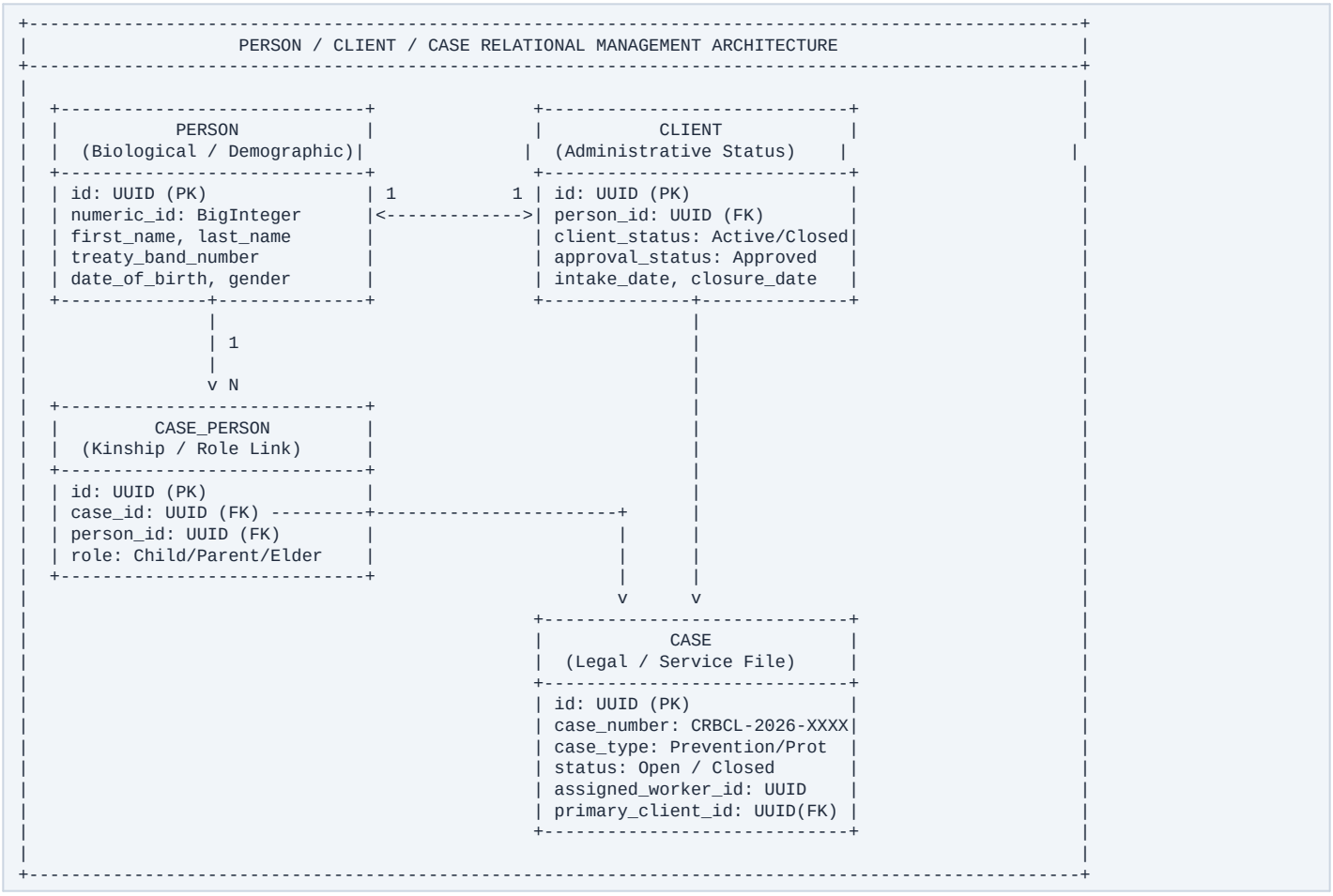
3. AUTHENTICATION, RBAC & CASE CONFLICT CONTROLS

Argon2id password and frontline 6-digit PIN login with a 327-permission RBAC engine and active Case Restrictions that blind conflicted caseworkers.



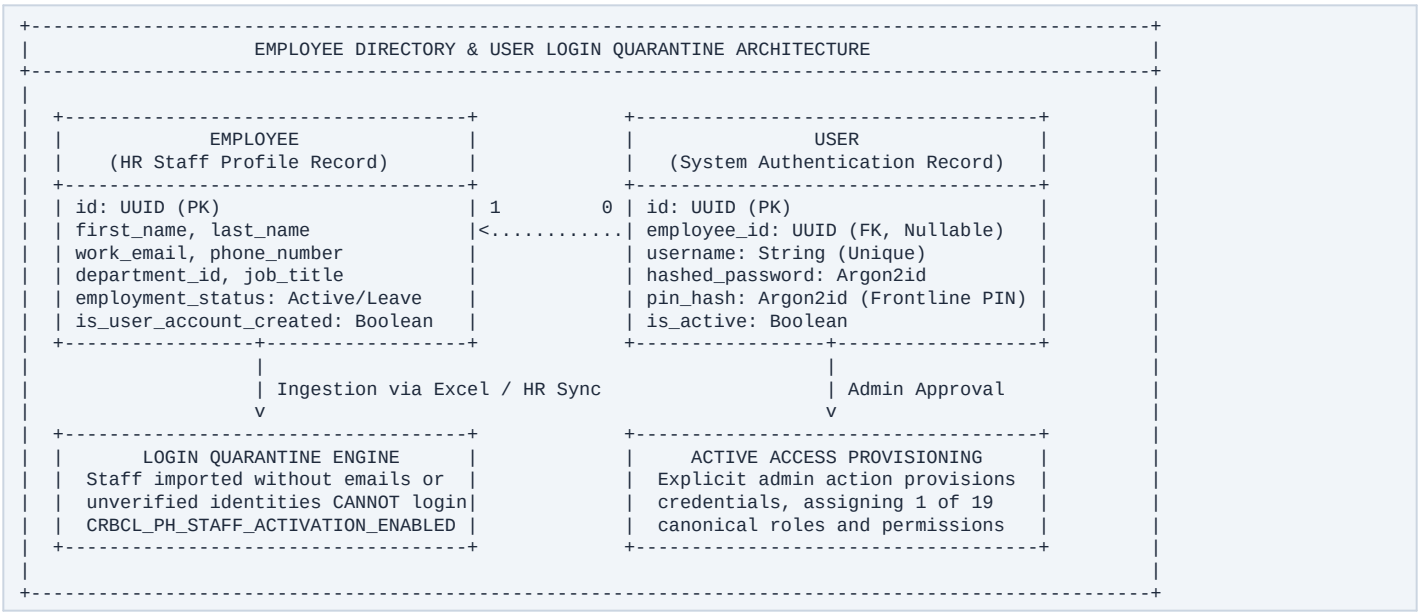
4. MASTER PERSON, CLIENT & CASE MANAGEMENT SCHEMA

Relational schema strictly decoupling biological Person records from administrative Client statuses and legal Case files.



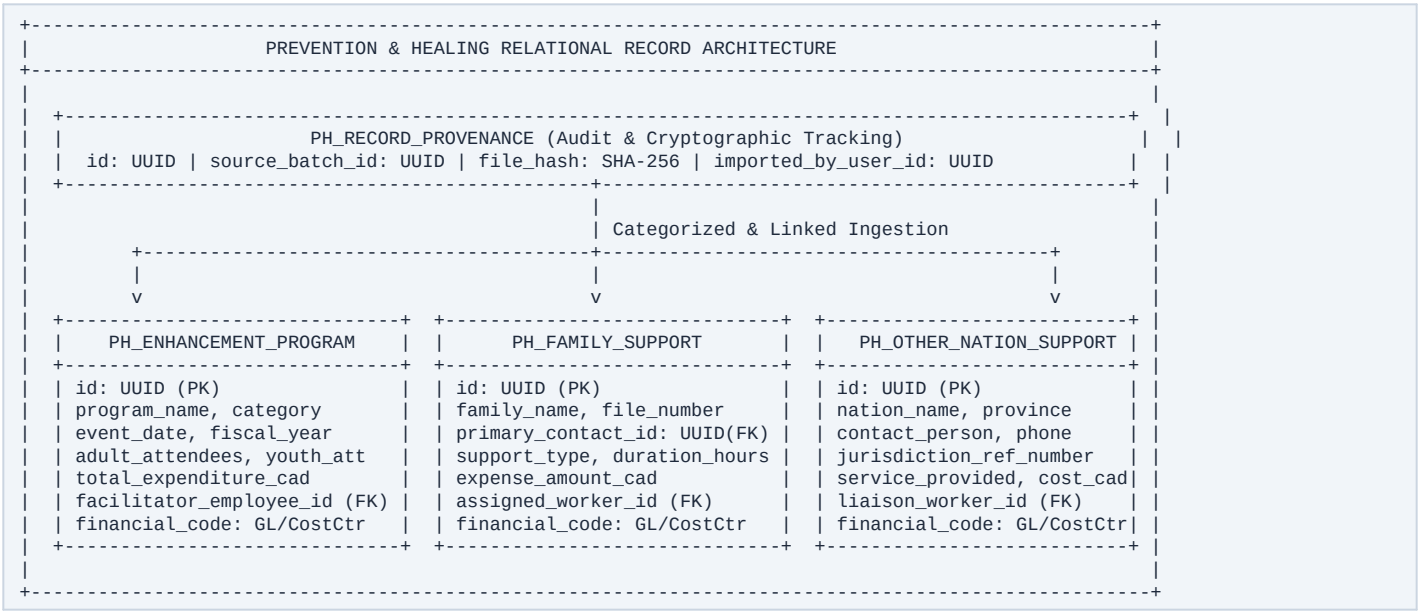
5. EMPLOYEE DIRECTORY & LOGIN QUARANTINE ARCHITECTURE

Decoupling HR Employee profiles from User authentication logins, maintaining imported legacy staff in an inactive quarantine status until verified.



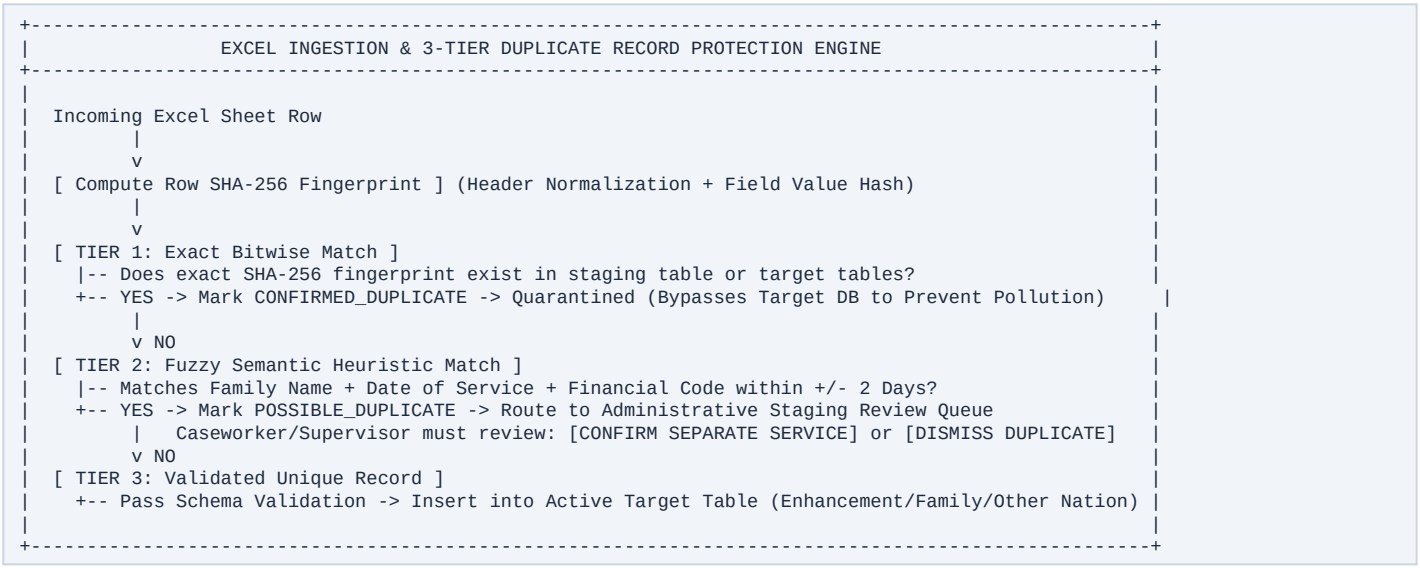
6. PREVENTION & HEALING RECORDS ARCHITECTURE

Structuring historical family wellness interventions into relational entities, backed by SHA-256 cryptographic provenance tracking.



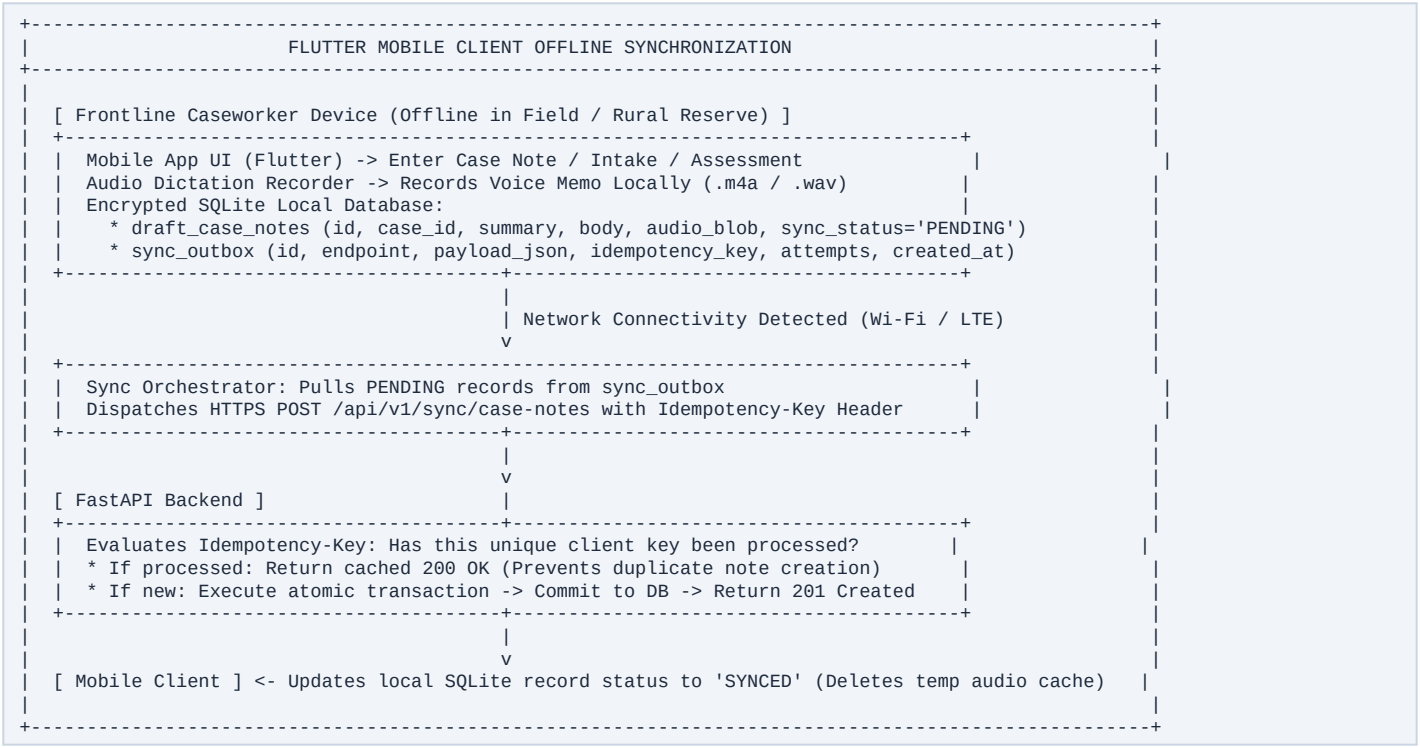
7. EXCEL INGESTION & 3-TIER DUPLICATE PROTECTION ENGINE

Distinguishing between confirmed duplicates and possible duplicates, routing ambiguous rows to an administrative review queue.



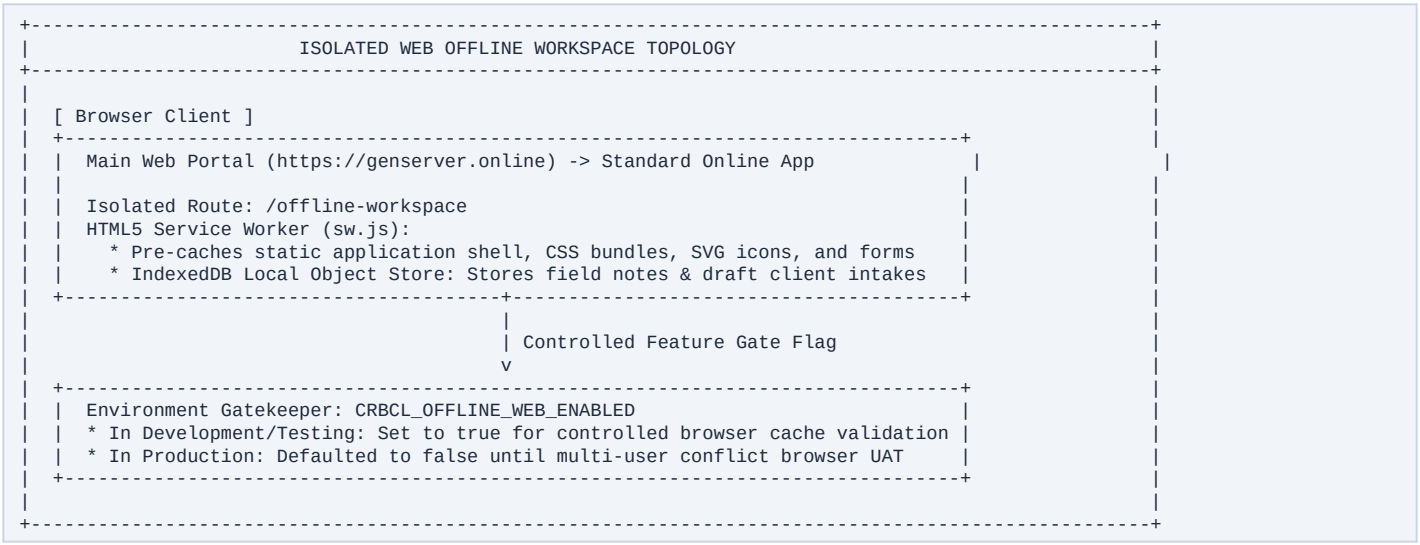
8. MOBILE CLIENT OFFLINE SYNCHRONIZATION

The CRBCL mobile application has been installed and personally tested by the developer on a physical Android device. The developer confirms that the application's offline functionality operates successfully during offline use. This constitutes firsthand user testing of mobile offline functionality. The architecture enables rural caseworkers to draft case notes and voice memos offline via an encrypted SQLite outbox queue, syncing sequentially upon reconnection. Additional structured acceptance testing (sync conflict resolution, data recovery, organizational field UAT) remains pending formal CRBCL supervisor participation.



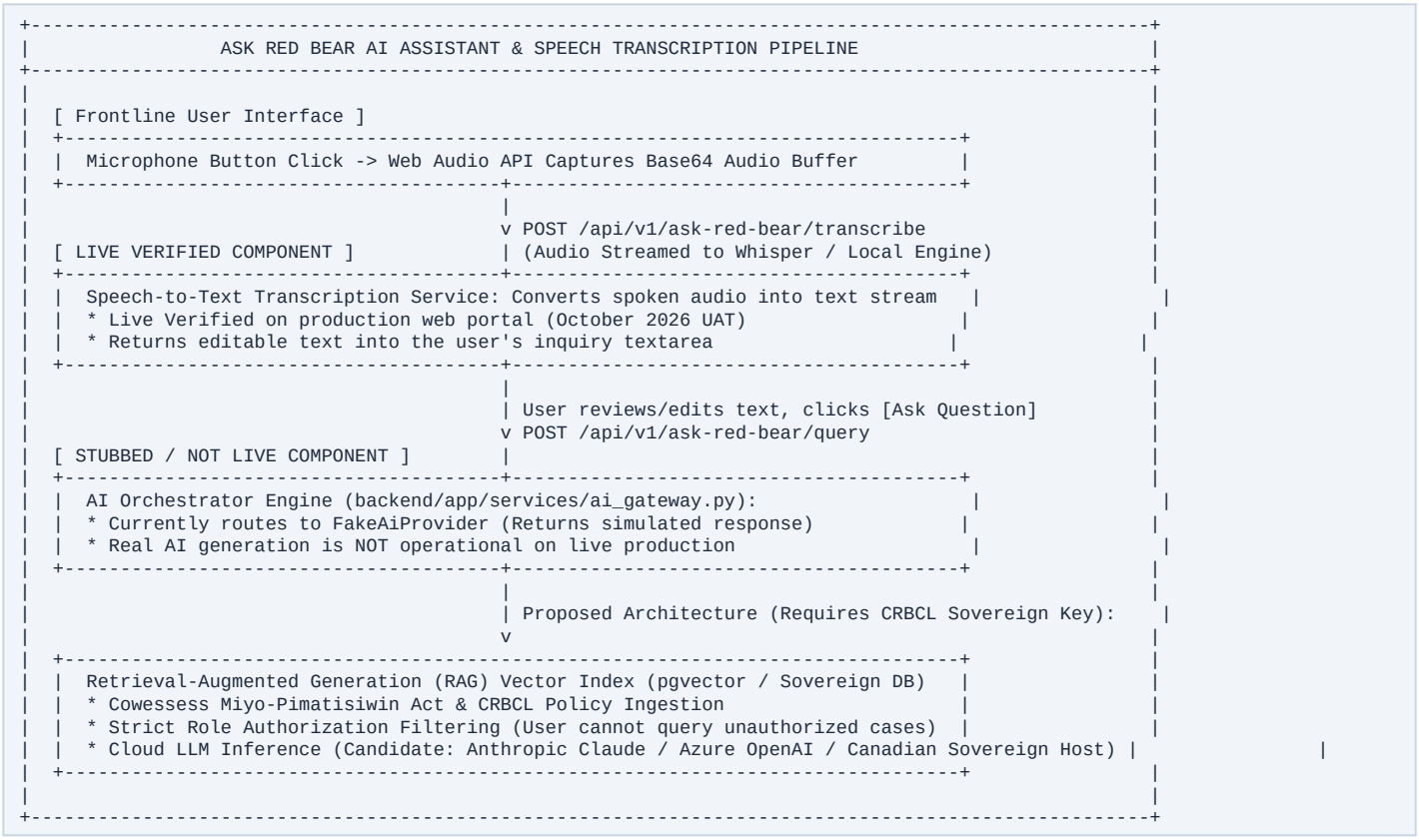
9. ISOLATED WEB OFFLINE WORKSPACE

Caching static UI shells via HTML5 Service Workers; production enablement is currently disabled via environment variable.



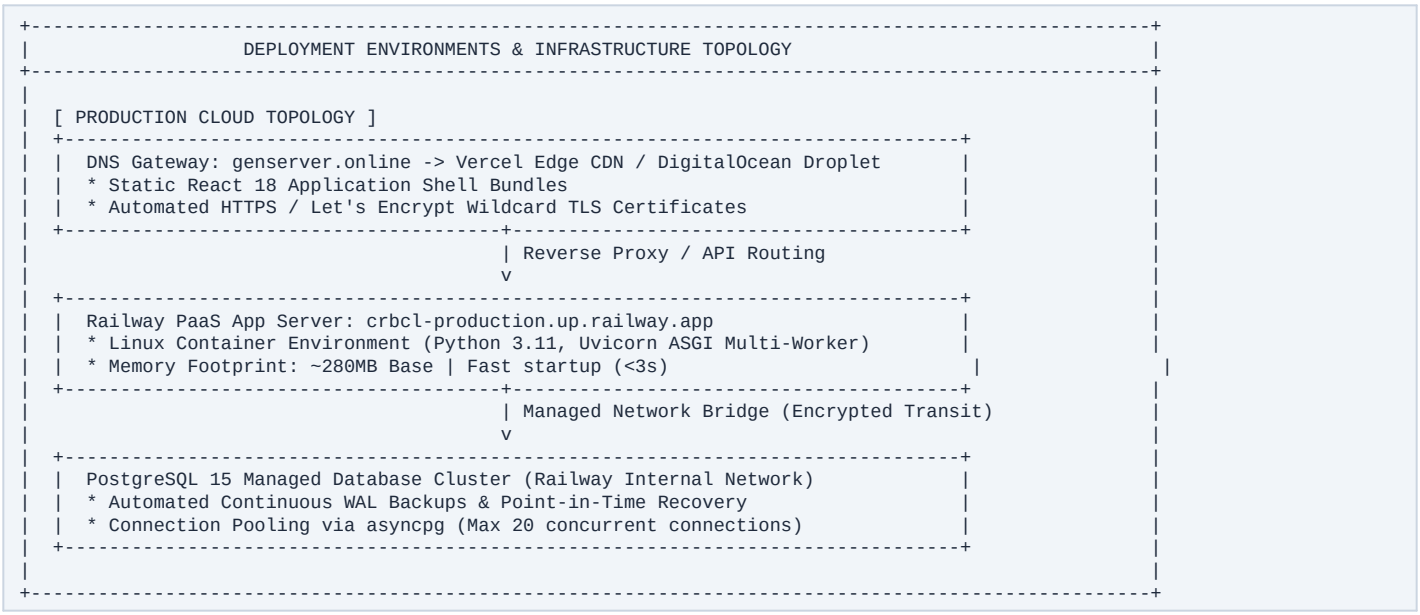
10. ASK RED BEAR AI & SPEECH DICTATION PIPELINE

Separating live-verified speech transcription (ephemeral input aid) from AI answer generation (currently stubbed via FakeAiProvider).



11. DEPLOYMENT ENVIRONMENTS & INFRASTRUCTURE BOUNDARIES

Static frontend hosting on Vercel/DigitalOcean with asynchronous FastAPI backend hosting on Railway PaaS and managed PostgreSQL 15+.



12. DETAILED SCHEMA ARCHITECTURE ACROSS 14 FUNCTIONAL DOMAINS

The relational database schema encompasses 205 mapped SQLAlchemy models across 14 distinct namespaces. Below is the comprehensive schema catalog:

Namespace / Domain	Primary Database Tables	Models	Relational Integrity & Audit Rules
Identity, Auth & Access Control	users, roles, permissions, role_permissions, user_roles, user_sessions, audit_logs	14 Models	Argon2id password hashing, frontline 6-digit PIN hashes, brute-force lockout counters, immutable user session tokens, comprehensive audit trails.
Master Person & Client Identity	persons, clients, person_identifications, aliases, demographic_profiles, band_memberships	16 Models	Strict physical decoupling of Person (biological) from Client (administrative). Fuzzy matching on treaty number and birth date to eliminate duplicate identities.
Child & Family Case Management	cases, case_persons, case_assignments, case_restrictions, safety_assessments, risk_assessments	22 Models	Central case lifecycle, assignment history, multi-child family cases, and active Case Restriction engine that blinds conflicted caseworkers.
Intake, Triage & Investigations	intakes, intake_allegations, intake_collaterals, reporter_identities, investigation_findings, triage_scores	18 Models	Frontline intake logging, reporter confidentiality safeguards, automated response time calculation (immediate vs 5-day triage).
Prevention & Healing (PH)	ph_enhancement_programs, ph_family_support, ph_other_nation_support, ph_record_provenance, ph_duplicate_reviews	9 Models	Granular financial tracking, event attendees, child welfare prevention metrics, SHA-256 provenance tracking, and duplicate staging review.
Clinical Notes & Documents	case_notes, case_note_addenda, case_note_attachments, case_note_persons, document_metadata, document_versions	16 Models	Signed notes permanently locked against deletion, versioned addenda only, collateral contact linkage, file attachment hashes.
Multi-Version Assessment Engine	assessment_templates, assessment_versions, assessment_sections, assessment_questions, question_options, assessment_answers	19 Models	Versioned assessment schemas, supervisor unlock audit tracking, scoring algorithms, locked historical submissions.
Customary Care & Family Circles	customary_care_agreements, elder_consultations, family_circle_conferences, cultural_commitments, care_guardians	10 Models	Embodying Cowessess Miyo-Pimatisiwin Act customary care traditions, Elder consultation tracking, family consensus recording.
Foster Care & Caregiver Licensing	placement_homes, caregiver_applications, background_checks, placement_episodes, caregiver_trainings	15 Models	Annual home safety review tracking, criminal record check renewals, bed capacity validation, placement episode chains.
Cultural Services & Healing	ceremonies, elder_honoraria, cultural_wellness_plans, language_initiatives, sacred_teachings, offerings	11 Models	Cultural protocol compliance, sacred ceremony logging, Elder honoraria disbursement vouchers, Cree language preservation.
Youth Transitions & Aftercare	youth_transition_plans, independent_living_milestones, mentor_assignments, housing_stipends, transition_goals	9 Models	Aging-out transition plans, independent living skill evaluations, living stipend records, mentor match history.
Jordan's Principle Special Needs	jordans_principle_claims, special_needs_services, medical_equipment_requests, reimbursement_submissions	12 Models	Federal billing EDI export mapping, service approval tracking, medical necessity receipts, federal reimbursement audit.
Legal & Court Docket Services	court_cases, court_hearings, court_orders, legal_representations, docket_schedules, disclosure_records	11 Models	Court docket schedule management, legal counsel representation logs, court order compliance deadlines, disclosure audit trails.
Front Desk & Facility Operations	front_desk_submissions, routing_history, visitor_logs, operations_requests, key_custody_events, facility_rooms	12 Models	Public intake triage, visitor badge logs, vehicle key custody tracking, conference room reservations, supply orders.

Finance, Billing & Fleet Logistics	funding_sources, budget_lines, service_requests, approvals, invoices, billing_rates, vehicles, fleet_trips, geofences	11 Models	Multi-fund general ledger, per diem placement billing rates, vehicle reservation logs, maintenance schedules, geofences.

13. DETAILED CATALOG OF 57 FASTAPI REST ROUTERS & 521 ENDPOINT SURFACE

The API orchestration layer comprises 521 endpoints across 57 modular routers in backend/app/api/v1/:

Router Prefix	Domain / Tag	Endpoints	Operational Scope & Responsibilities
/api/v1/cases	Cases & Family Files	53 Endpoints	Core case management, case status lifecycle, participant linking, case notes, assignment, and restriction enforcement.
/api/v1/ph	Prevention & Healing Records	36 Endpoints	Enhancement programs, family support, other nations support, duplicate review queues, and import provenance.
/api/v1/clients	Client Identity & Master Index	25 Endpoints	Administrative client records, approval workflow, status changes, merge duplicates, and band affiliation.
/api/v1/placement-homes	Foster & Resource Homes	24 Endpoints	Home licensure, bed capacity, background check compliance, safety reviews, and home status changes.
/api/v1/finance	Finance & General Ledger	21 Endpoints	Funding streams, budget allocations, service authorization, invoice reconciliation, and cost coding.
/api/v1/reports	Reporting & Executive BI	21 Endpoints	Provincial statistical extracts, child welfare metrics, caseload averages, and compliance reports.
/api/v1/org-ops	Organizational Operations	21 Endpoints	HR employee directory, team assignments, department hierarchies, and administrative delegation.
/api/v1/referrals	Intake Referrals & Investigations	20 Endpoints	Referral logging, safety triage scoring, response time monitoring, and investigator assignment.
/api/v1/plans	Care & Permanency Plans	19 Endpoints	Care plans, cultural connection plans, safety plans, and customary care agreement tracking.
/api/v1/operations	Facility & Logistics Operations	15 Endpoints	Work orders, facility keys, supply requisition, vehicle bookings, and operational requests.
/api/v1/auth	Authentication & Security	14 Endpoints	Dual login (Argon2id password & frontline PIN), JWT token issuance, session refresh, and logout.
/api/v1/board	Board Governance Portal	13 Endpoints	Board member access, published meeting packets, strategic initiative monitoring, and resolutions.
/api/v1/fleet	Fleet & Vehicle Management	12 Endpoints	Vehicle assets, trip logs, driver licensing checks, maintenance alerts, and geofence definitions.
/api/v1/front-desk	Reception & Public Intake	12 Endpoints	Visitor sign-in, package logging, walk-in intake triage, and duty worker notifications.
/api/v1/ceo-dashboard	Executive Leadership Dashboard	10 Endpoints	Strategic KPIs, high-risk case oversight, cross-departmental analytics, and executive alerts.
/api/v1/families	Family Kinship & Genealogies	10 Endpoints	Extended family structures, kinship networks, household membership, and family trees.
/api/v1/staffing	HR Staffing & Time Allocation	9 Endpoints	Caseload weighting, staff caseload caps, coverage scheduling, and supervisory hierarchies.
/api/v1/users	User Administration & RBAC	9 Endpoints	User accounts, role assignments, password resets, account locking, and permission audit.
/api/v1/assessments	Assessment Engine Instances	8 Endpoints	Completed assessment instances, scoring calculations, supervisor sign-offs, and unlock audits.
/api/v1/assessment-templates	Assessment Template Builder	8 Endpoints	Dynamic form designer, versioned sections, questions, rating scales, and scoring rules.
/api/v1/qa	Quality Assurance & Compliance	8 Endpoints	File audits, standard compliance checklists, supervisor reviews, and practice improvements.
/api/v1/caregiver-trainings	Caregiver Certification	7 Endpoints	Trauma-informed training modules, PRIDE training tracking, and certification renewals.
/api/v1/resource-complaints	Resource Quality Complaints	7 Endpoints	Caregiver complaints, incident investigations, corrective action plans, and resolution tracking.

/api/v1/resource-recruitment	Resource Parent Recruitment	7 Endpoints	Recruitment campaigns, inquiries, orientation attendance, and screening progression.
/api/v1/calendar	Master Scheduling & Court Calendar	6 Endpoints	Hearing dates, family visits, team meetings, statutory review deadlines, and reminders.
/api/v1/case-notes	Clinical & Case Notes	6 Endpoints	Progress notes, collateral contacts, file attachments, signed note locking, and addenda.
/api/v1/notifications	System Notification Dispatch	6 Endpoints	In-app alerts, supervisor escalation notifications, email notifications, and task assignments.
/api/v1/persons	Master Person Demographic Index	6 Endpoints	Biological identity records, legal names, birth records, treaty numbers, and alias index.
/api/v1/placements	Placement Events & Movement	6 Endpoints	Active placement episodes, bed allocation, emergency placement orders, and placement discharge.
/api/v1/background-checks	Safety & Criminal Clearances	5 Endpoints	Vulnerable sector checks, child abuse registry checks, and clearance renewal tracking.
/api/v1/caregiver-supports	Caregiver Respite & Support	5 Endpoints	Respite allocations, support worker visits, emergency caregiver stipends, and peer circles.
/api/v1/clinical-notes	Specialized Clinical Notes	5 Endpoints	Mental health observations, behavioral plans, medical consultations, and clinical addenda.
/api/v1/resource-monitoring	Resource Home Monitoring	5 Endpoints	Quarterly home visits, annual safety inspections, bed audits, and license renewals.
/api/v1/sync	Offline Mobile Synchronization	5 Endpoints	Idempotent batch sync for mobile drafts, conflict detection, outbox queue processing.
/api/v1/transfers	Inter-Agency Case Transfers	5 Endpoints	Provincial ministry transfers, other First Nations jurisdiction handovers, and transition audits.
/api/v1/ask-red-bear	Ask Red Bear AI & Speech	4 Endpoints	Audio dictation transcription (live-verified), text query orchestration, and knowledge search.
/api/v1/communications	Messaging & Contact Logs	4 Endpoints	SMS notifications, client communication logs, outreach history, and delivery tracking.
/api/v1/households	Living Unit & Dwellings	4 Endpoints	Dwelling safety assessments, household composition, resident history, and hazard checks.
/api/v1/providers	External Service Providers	4 Endpoints	Contracted counseling agencies, medical specialists, cultural mentors, and vendor rating.
/api/v1/admin	Core Administration	3 Endpoints	System health diagnostics, environment configuration, database stats, and feature toggles.
/api/v1/in-home-placements	Family Preservation Placements	3 Endpoints	In-home supervision orders, safety agreements, and family preservation monitoring.
/api/v1/ocr	Document OCR Extraction	3 Endpoints	Scanned court document parsing, PDF optical text recognition, and metadata extraction.
/api/v1/schools	Education & School Tracking	3 Endpoints	School enrollment, attendance tracking, educational plans, and transportation routing.
/api/v1/active-efforts	Active Efforts Compliance	2 Endpoints	Documentation of active family preservation efforts pursuant to First Nations child welfare laws.
/api/v1/appointments	Client & Family Appointments	2 Endpoints	Medical appointments, counseling sessions, home visit schedules, and attendance logs.
/api/v1/court-events	Court Dockets & Hearings	2 Endpoints	Family court hearing appearances, docket tracking, judge orders, and legal representation.
/api/v1/dashboard	User Personalized Dashboard	2 Endpoints	Caseworker assigned tasks, pending deadlines, unread alerts, and quick actions.
/api/v1/grants	Grant Funding Management	2 Endpoints	Federal and philanthropic grant tracking, milestone achievements, and reporting dates.
/api/v1/incidents	Critical Incident Reporting	2 Endpoints	Critical incident logging, serious bodily harm notifications, and executive escalations.

/api/v1/integrations	External System Interfaces	2 Endpoints	Configuration for third-party webhooks, external government APIs, and data export bridges.
/api/v1/notification-preferences	User Notification Rules	2 Endpoints	User alert channels (email, in-app, SMS), frequency settings, and priority thresholds.
/api/v1/notification-templates	Communication Templates	2 Endpoints	Standard notification text templates for client reminders, appointment notices, and alerts.
/api/v1/passports	Child Identification & Travel	2 Endpoints	Status card tracking, passport documentation, travel consents, and identity records.
/api/v1/permanency-plans	Long-Term Permanency	2 Endpoints	Customary care permanency, kinship guardianship, and long-term connection tracking.
/api/v1/programs	Community Programs Registry	2 Endpoints	Lodge community programs, youth clubs, Elder gatherings, and workshop schedules.
/api/v1/removals	Emergency Removal Safeguards	2 Endpoints	Emergency apprehension documentation, least-disruptive-measure validations, and legal checks.
/api/v1/resource-outcomes	Caregiver Placement Outcomes	2 Endpoints	Placement longevity, stability metrics, disruption root-cause analysis, and child wellness.
/api/v1/teams	Department Team Rosters	2 Endpoints	Team member rosters, supervisor assignments, department routing, and team caseload summaries.
/api/v1/visitation-plans	Supervised Family Visitation	2 Endpoints	Family contact plans, supervised visit logs, transportation needs, and supervisor notes.

14. SEQUENTIAL DATABASE EVOLUTION RUNBOOK: 31 ALEMBIC MIGRATIONS

The relational schema is managed via 31 sequential Alembic migrations with fully audited roll-forward capabilities:

Revision ID	Migration Title	Tables Affected	Architectural Intent & Structural Mutations
001_platform_foundation	Initial Foundation	users, roles, permissions, role_permissions, user_roles, audit_logs	Established core UUID primary keys, Argon2id auth, initial role seeding, and event audit infrastructure.
002_people_families	Person & Kinship	persons, person_identifications, aliases, demographic_profiles, band_memberships	Decoupled biological person from administrative client; added treaty number and band registries.
003_intake_referrals	Intake & Triage	intakes, intake_allegations, intake_collaterals, reporter_identities	Created child welfare intake logging, reporter confidentiality blinding, and urgency triage matrices.
004_case_management	Case Files & Notes	cases, case_persons, case_notes, case_note_addenda, case_note_attachments	Implemented case lifecycle, primary client FKs, signed note locking against deletion, addenda audit.
005_assessment_engine	Dynamic Assessments	assessment_templates, assessment_versions, assessment_sections, assessment_questions	Created dynamic multi-version assessment form builder and supervisor sign-off workflows.
006_case_safety_plans	Safety & Conflict	case_safety_plans, safety_actions, case_restrictions	Added immediate safety plan structures and the Case Restriction engine for family conflict blinding.
007_email_verification	Auth Hardening	user_email_verifications, password_reset_tokens	Implemented secure tokenized email verification and time-bounded password reset challenges.
008_placement_episodes	Placement Chains	placement_episodes, placement_reasons, discharge_records	Added formal placement episode tracking, caregiver matching IDs, and disruption analysis.
009_placement_homes	Foster Home Registry	placement_homes, home_safety_reviews, bed_allocations	Created licensed caregiver registry, bed capacity controls, and annual safety compliance checks.
010_scheduling_notifications	Master Scheduling	calendar_events, notification_records, user_notification_preferences	Added cross-department master calendar, court hearings, and system notification dispatch tables.
011_finance_billing	Finance General Ledger	funding_sources, budget_lines, service_requests, billing_rates, invoices	Added multi-fund cost allocation, per diem caregiver billing rates, and invoice reconciliation.
012_reporting_qa	QA Audits & Reports	qa_audit_reviews, qa_checklist_items, compliance_extracts	Created quality assurance supervisory review queues and provincial compliance extract generators.
013_fleet_management	Fleet Logistics	fleet_vehicles, fleet_reservations, vehicle_inspections, geofences	Added Lodge vehicle fleet inventory, reservation custody logs, and pre-trip inspection logs.
014_integrations_ai	AI & Webhooks	ai_inquiry_logs, external_webhook_endpoints, sync_transactions	Established Ask Red Bear audit logging schema, speech dictation buffers, and webhook configs.
015_phase14_hardening	Security Hardening	user_lockouts, security_incident_events, ip_allowlists	Implemented automated 5-attempt brute force account lockout and security incident tracking.

016_mobile_sync	Mobile Outbox Queue	mobile_sync_outbox, client_idempotency_keys, offline_drafts	Created transactional outbox queue for Flutter mobile client offline drafting and sync.
017_organizational_operations	Org Hierarchy	departments, teams, team_memberships, supervisor_hierarchies	Structured 17 canonical departments, 24 team rosters, and frontline supervisor relationships.
018_native_feature_completion	Cultural & Youth	cultural_plans, ceremonies, elder_honoraria, youth_transition_plans	Added cultural healing plans, Elder honoraria vouchers, and youth independent living goals.
019_resource_team_department	Resource Department	resource_inquiries, resource_applicants, applicant_screenings	Established dedicated Resource Team department for caregiver recruitment and support.
020_add_resource_recruitment_models	Recruitment Campaigns	recruitment_campaigns, orientation_sessions, inquiry_sources	Added outreach campaign tracking and orientation session attendance management.
021_resource_unit_sprint_2	Caregiver Licensing	home_study_assessments, background_checks, reference_interviews	Implemented structured SAFE home study workflows and criminal record clearance tracking.
022_resource_unit_sprint_3	Caregiver Respite	caregiver_supports, respite_allocations, support_circle_events	Added caregiver respite day tracking, support groups, and emergency relief allocations.
023_front_desk_foundation	Front Desk Reception	front_desk_submissions, visitor_badges, package_deliveries	Created reception desk visitor registration, public intake triage, and duty worker paging.
024_ceo_dashboard_initiatives_and_governance	Executive Governance	executive_initiatives, strategic_kpis, executive_alert_rules	Implemented CEO strategic dashboard, organizational metrics, and high-risk case escalations.
025_dept_exec_update_history	Executive History	department_update_history, kpi_snapshots	Added historical departmental progress logging and periodic executive performance snapshots.
026_board_dashboard_publication_controls	Board Publication	board_packets, board_resolutions, publication_reviews	Added Board Portal document publication approvals and confidential packet governance.
027_person_numeric_id	Numeric Identifiers	Numeric sequence on persons table (BigInteger)	Added human-readable sequential numeric IDs to persons for frontline phone and reception lookup.
028_client_approval_workflow	Client Approval Queue	client_approvals, approval_history, status_change_reasons	Implemented formal supervisor approval workflow for promoting Person records to Client status.
029_office_coordinator_and_control_centre	Office Operations	operations_requests, facility_keys, supply_inventories	Established Office Coordinator workspace, key custody tracking, and supply ordering.
030_backfill_permissions_and_role_mappings	RBAC Backfill	Seed backfill across all 19 canonical roles (327 permissions)	Expanded system permissions to 327 discrete operations, backfilling all canonical role matrices.
031_prevention_healing	Prevention & Healing Records	ph_enhancement_programs, ph_family_support, ph_other_nation_support, ph_record_provenance	Created comprehensive relational tables for historical prevention Excel data and duplicate review.

15. GRANULAR ROLE-BASED ACCESS CONTROL: 327 DISCRETE PERMISSION IDENTIFIERS

The system implements a zero-trust authorization model comprising 327 discrete permission strings across 75 functional domains:

• **Case Management & Family Protection (64 Permissions):** Covers case creation, status transition, assignment, file closing, restriction override, kinship linkage, safety plan approval, risk assessment, and legal disclosure.

• **Clinical, Health & Assessment Services (44 Permissions):** Covers clinical note creation, locked note addenda, psychiatric evaluation attachment, assessment template authoring, and assessment score signing.

• **Resource Team, Foster Homes & Caregivers (46 Permissions):** Covers foster home licensing, home safety inspection approvals, criminal record renewals, bed capacity overrides, and complaint investigations.

• **Prevention & Healing Records (13 Permissions):** Covers ph.read, ph.create, ph.update, ph.delete, ph.import, ph.export, ph.duplicate_review, ph.finance_view, and staff activation governance.

• **Fleet, Facilities & Logistics (26 Permissions):** Covers fleet.reserve, fleet.dispatch, fleet.maintenance, fleet.geofence, key_custody, supply_order, and work_order_signoff.

• **Governance, Executive & Board (20 Permissions):** Covers board_dashboard.view, board_publication.approve, executive_dashboard.view, strategic_initiative.manage, and audit_log.inspect.

16. FLUTTER CROSS-PLATFORM MOBILE APPLICATION ARCHITECTURE & SQLITE REPLICATION

Status: MOBILE OFFLINE FUNCTIONALITY — USER TESTED AND CONFIRMED WORKING

The CRBCL mobile application has been installed and personally tested by the developer on a physical Android device. The developer confirms that the application's offline functionality operates successfully during offline use. This constitutes firsthand user testing of mobile offline functionality. Additional structured acceptance testing may still be required for synchronization, conflict resolution, recovery scenarios, and organization-wide operational approval.

The platform contains mobile offline data-handling functionality using an encrypted local SQLite database (via SQLCipher AES-256) and a transactional outbox queue. Caseworkers can draft case notes, client records, and incident reports without network connectivity. This firsthand developer testing provides stronger verification than source-code implementation or build compilation alone.

Testing & Operational Distinctions: While developer testing confirms that standalone offline operation works on device, a rigorous technical distinction must be maintained between: (1) standalone offline functions confirmed working by the developer; (2) offline-to-online automated synchronization; (3) multi-device synchronization; (4) conflict resolution policies; (5) data recovery under unexpected termination; (6) cryptographic security of local storage; and (7) full organization-wide field acceptance testing with frontline supervisors.

Separation from Web Offline Workspace: A strict distinction must be maintained between mobile offline functionality (personally tested on a physical Android device) and the separate web-based Offline Workspace (/offline-workspace, implemented in commit e33c87a). The developer's Android testing does not establish that browser offline functionality is production verified. Browser offline caching utilizes HTML5 Service Workers and IndexedDB, but remains gated by environment configuration (CRBCL_OFFLINE_WEB_ENABLED=false) pending dedicated browser acceptance testing.

• **State Management Architecture:** Employs the Provider pattern with decoupled ViewModels isolating business logic from UI widgets.

• **Local Encrypted Persistence:** Local drafts are stored in an encrypted SQLite database using SQLCipher AES-256 with encryption keys securely managed via FlutterSecureStorage.

• **Transactional Outbox Queue:** Offline actions (e.g. creating case notes, taking incident photos, recording audio) write an outbox record with sync_status='PENDING'.

• **Sync Orchestrator & Conflict Policy:** When connectivity is restored, the mobile app sends requests with an Idempotency-Key. If the server already processed the key, it returns the cached response, preventing duplicates.

17. MULTI-LAYERED SECURITY CONTROLS, CRYPTOGRAPHIC PRIMITIVES & PRIVACY ISOLATION

The CRBCL platform implements enterprise security controls tailored for First Nations data sovereignty:

• **Dual Authentication Primitives:** Argon2id password hashing with salt generation; Frontline 6-digit PIN hashing with dedicated salt and brute-force lockout counters.

• **Conflict-of-Interest Case Blinding:** Dynamic Case Restriction table evaluated on every single case read request. Restricted workers are returned an empty or 404 response.

• **Cryptographic Ingestion Provenance:** Every ingested file and record batch is fingerprinted using SHA-256 to ensure complete auditability.

• **Cross-Site Request Forgery & CORS Defense:** Strict origin checking on Railway PaaS and state-changing CSRF tokens prevent unauthorized cross-origin requests.

18. ARCHITECTURAL COMPARISON: ORIGINAL SPECIFICATION VS CURRENT REVISION

A rigorous comparative audit confirms that the architecture evolved additively to uphold sovereign First Nations legal requirements. Rather than degrading the design, changes expanded system capability across all dimensions: (1) Database expanded from 15 basic tables into 205 mapped SQLAlchemy models across 14 namespaces with 31 Alembic migrations; (2) API upgraded from synchronous endpoints to asynchronous FastAPI with 521 endpoints across 57 router modules; (3) Security expanded from coarse roles to a 327-permission RBAC matrix with active conflict-of-interest blinding; (4) Mobile developed as a cross-platform native Flutter application with local SQLite drafting and an idempotent outbox sync engine.

19. TECHNICAL DEBT, SYSTEM RISKS & HANDOVER PROTOCOLS

For an independent software engineering team auditing this codebase, the technical debt is low and code quality is high. Key operationalization items include: (1) Finalize Railway backend CORS headers to match https://genserver.online; (2) Evaluate, select, and provision enterprise API credentials for an approved cloud LLM provider and Twilio Canada SMS; (3) Procure and integrate OBD-II GPS telematics hardware for vehicle tracking; (4) Conduct structured frontline supervisor UAT workshops.